

5. Project Development Phase

ComicCraft - AI/ML & Generative AI Project

1. Implementation Summary

ComicCraft is implemented as a web-based Generative AI application. The application combines a FastAPI backend, a browser-based frontend, and Gemini AI to convert a user's creative prompt into a structured comic story.

2. Development Components

- FastAPI backend for routing and request handling.
- HTML, CSS and JavaScript frontend for user interaction.
- Gemini AI integration for Generative AI content creation.
- Environment-based configuration for API credentials.
- Prompt and response processing for structured comic output.

3. Application Workflow

Step	Process
1	User enters a creative comic/story idea.
2	Frontend sends the prompt to the backend generation API.
3	FastAPI receives and processes the request.
4	Gemini AI generates the comic story content.
5	Backend returns the generated response.
6	Frontend displays the generated story and four-panel structure.

4. Backend Development

The FastAPI backend provides the application routes and handles the communication between the web interface and the Generative AI service. It is responsible for receiving the prompt, processing the request, calling the AI generation layer, and returning the result.

5. Frontend Development

The frontend provides a simple interface for entering a prompt and viewing the generated comic. HTML structures the page, CSS handles presentation, and JavaScript manages user interaction and API requests.

6. Generative AI Integration

Gemini AI is used as the Generative AI component. The prompt is passed to the AI service with instructions to create a coherent comic-style story and organize the result into four panels.

7. Configuration and Security

Sensitive configuration such as API credentials is intended to be stored in environment variables rather than directly inside source code. This helps keep credentials separate from the application source.

8. Current Working Version

The corrected implementation is maintained as **ComicCraft_Corrected_Project**. The development baseline includes the working FastAPI application, frontend flow, and AI-generation integration.

9. Development Outcome

- Working web application foundation.
- Connected frontend and backend request flow.
- Gemini AI generation integration.
- Structured four-panel comic text output.
- Architecture prepared for future visual comic generation.

10. Future Development

- Generate images for each comic panel.
- Maintain character appearance across panels.
- Add speech bubbles and improved dialogue presentation.
- Provide PDF and image download options.
- Add comic style and character customization controls.